of which cannot be accurately foretold is called annuity contingent.

For example the endowment funds of trust, where the interest earned is used for welfare activities only. The principal remains the same and activity continues forever.

All the above types of annuities are based on the number of their periods. An annuity can also be classified on the basis of mode of payment as under.

b) Based on the mode of payment :

i) **Ordinary annuity:** An annuity in which payments of installments are made at the end of each period is called ordinary annuity (or immediate annuity)

For example repayment of housing loan, vehicle loan etc.,

ii) **Annuity due:** An annuity in which payments of installments are made in the beginning of each period is called annuity due.

In annuity due every payment is an investment and earns interest. Next payment will earn interest for one period less and so on, the last payment will earn interest of one period.

For example saving schemes, life insurance payments, etc.



Deferred annuity: An annuity which is payable after the lapse of a number of periods is called deferred annuity.

The derivation of the following formulae are given for better understanding and are exempted from examination.

(i) Amount of immediate annuity (or) Ordinary annuity (or) Certain annuity

Let 'a' be the ordinary annuity and i percent be the rate of interest per period. In ordinary annuity, the first installment is paid after the end of first period. Therefore it earns interest for $(n-1)$ period, second installment earns interest for $(n-2)$ periods and so on. The last installment earns for $(n-n)$ periods. (i.e) earns no interest.

For $(n-1)$ periods,

The amount of first annuity = $a(1+i)^{n-1}$

The amount of second annuity = $a(1+i)^{n-2}$

The amount of third annuity = $a(1+i)^{n-3}$

and so on.

$\therefore$ The total amount of annuity A for n period at i percent rate of interest is

$A = a(1+i)^{n-1}+a(1+i)^{n-2}+...a(1+i)+a$

$= a[(1+i)^{n-1}+(1+i)^{n-2}+...+(1+i)+1]$

$= a[1+(1+i)+(1+i)^2+...+(1+i)^{n-1}]$

$= a[1+r+r^2+...+r^{n-1}]$, where $1+i=r$

$= a[\frac{r^n-1}{r-1}]$, G.P with common ratio $r > 1$

$= a\left[\frac{(1+i)^n-1}{1+i-1}\right]$

$\boldsymbol{A = \frac{a}{i}[(1+i)^n-1]}$

(ii) Present Value of immediate annuity (or ordinary annuity)

Let 'a' be the annual payment of an ordinary annuity, n be the number of years and

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